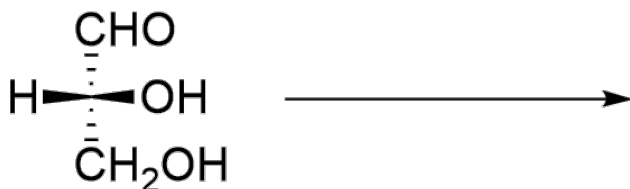
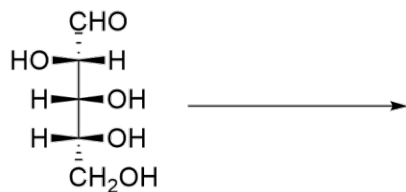
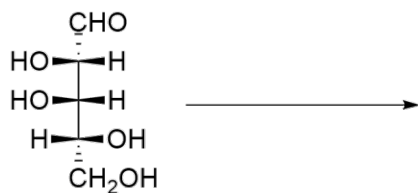
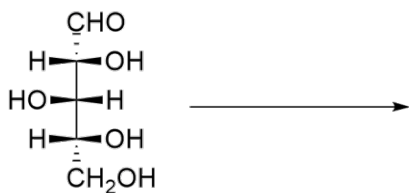
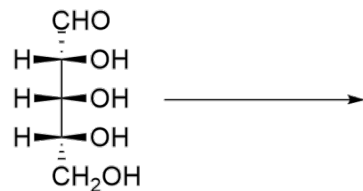


Fischer projections and Isomers

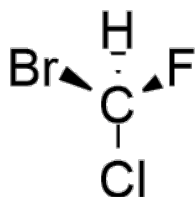
1. Define the term Fischer projection.
2. Convert the following molecule (R)-Glyceraldehyde to a Fischer projection.



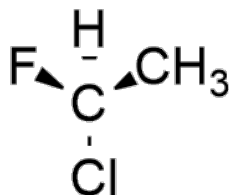
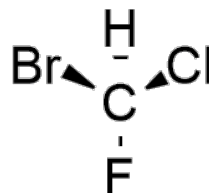
3. Convert the following 3-D representation molecules to Fischer projections.



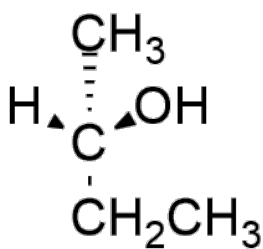
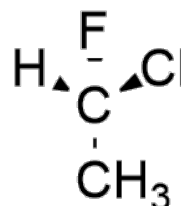
- Write the enantiomeric forms of bromochlorofluoromethane and assign each enantiomer its correct (R) or (S) designation.
- Tell whether the two structures in each pair represent enantiomers or two molecules of the same compound in different orientations.



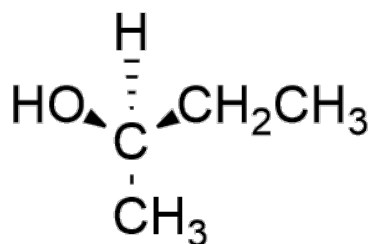
and



and

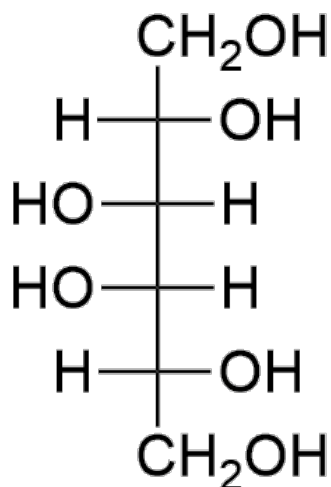


and





6. D-Galactitol is one of the toxic compounds produced by the disease galactosemia. Accumulation of high levels of D-Galactitol causes the formation of cataracts. A Fischer projection for D-Galactitol is shown here.



- Draw the three dimensional structure for D-Galactitol.
- Draw the mirror image of D-Galactitol and write its Fischer projection formula.
- What is the stereochemical relationship between D-Galactitol and its mirror image?